Machine Learning - Undergraduate Level Assessment 6 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which image data augmentation is most common for natural images?
   (a) random crop and horizontal flip
   (b) random crop and vertical flip
   (c) posterization
   (d) dithering

2. As of 2020, which architecture is best for classifying high-resolution images?
   (a) convolutional networks
   (b) graph networks
   (c) fully connected networks
   (d) RBF networks

3. Averaging the output of multiple decision trees helps _.
   (a) Increase bias
   (b) Decrease bias
   (c) Increase variance
   (d) Decrease variance

4. In Yann LeCun's cake, the cherry on top is
   (a) reinforcement learning
   (b) self-supervised learning
   (c) unsupervised learning
   (d) supervised learning

5. Statement 1| A neural network's convergence depends on the learning rate. Statement 2| Dropout multiplies randomly chosen activation values by zero.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The joint probability P(H, U, P, W) can be expressed as P(H) * P(W) * P(P | W) * P(U | H, P).

7. True or False: The method used to learn weights, whether by matrix inversion or gradient descent, primarily affects underfitting and overfitting in polynomial regression.

8. True or False: As of 2020, certain models achieved over 98% accuracy on the CIFAR-10 dataset.

9. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as that of the Maximum Likelihood Estimate (MLE).

10. True or False: An excessively large step size can cause the training loss to increase over epochs during the learning process.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to shift last element to first position in the given list.

12. Write a function to find the maximum sum of bi-tonic sub-sequence for the given array.

End of Paper